TREEWALK NYC • STEAM LESSON PLAN

# Cherry

*Prunus serrulata (and related ornamental cultivars)*

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**Species background:** New York's street cherries are mostly cultivated ornamental varieties bred from Japanese and Asian wild cherry species, popularized worldwide after Japan gifted flowering cherries to Washington, D.C. in 1912. Look for horizontal dashes across smooth, reddish-brown bark -- these lenticels (breathing pores) are one of the clearest identifying marks across almost all ornamental cherry varieties.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

# Cherry: Find the tiny teeth on a cherry leaf's edge and the dashes on its bark.

**Standards connection:** NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

**Materials:** Coloring sheet, crayons.

## Teacher background

Find the tiny teeth along the edge of a leaf. Look for tiny dashes on the bark -- lenticels, like little pores.

## Procedure

1. Engage: Ask students to gently feel a leaf's edge (with care) and describe how it feels.
2. Explore: Look at the bark together and count the little horizontal dashes.
3. Explain: Explain that these dashes are lenticels, tiny breathing holes.
4. Art: Feel the tiny teeth on a leaf's edge, then draw a zigzag line to show them.
5. Evaluate: Student can point to the toothed leaf edge and the bark dashes.

## Discussion questions

What other things have a bumpy or toothed edge?

## Assessment

Correctly points out both features with prompting.

## Extension

Compare this tree's bark dashes to another tree's bark on your walk.

GRADE BAND 3-5

# Cherry: Compare cherry's lenticels to another species' and observe seasonal bloom.

**Standards connection:** NGSS 3-ESS2-1; National Core Arts Standards VA:Cr2 (3-5)

**Materials:** Coloring sheet, colored pencils.

## Teacher background

Compare cherry's lenticels to zelkova's -- same structure, different species, good side-by-side observation. In spring, cherries can be covered in pink or white flowers almost overnight.

## Procedure

1. Engage: Ask what it would look like if a tree changed appearance overnight.
2. Explore: If it's spring, observe how much of the tree is covered in blossoms.
3. Explain: Discuss lenticels as a shared bark feature across several unrelated species.
4. Art: Sketch a branch of blossoms using only pink and white paint dabbed on with a fingertip or cotton swab.
5. Evaluate: Student writes one sentence describing the tree's rapid spring change.

## Discussion questions

Why might it help a tree to bloom quickly, all at once?

## Assessment

Correct sentence describing the bloom's rapid, synchronized nature.

## Extension

Find another tree with lenticels on its bark and compare the pattern.

GRADE BAND 6-8

# Cherry: Explain bloom synchrony and compare bark gas-exchange structures across species.

**Standards connection:** NGSS MS-ESS3-5; National Core Arts Standards VA:Cr2/Re7 (6-8)

**Materials:** Coloring sheet, ink-wash or watercolor materials.

## Teacher background

Cherry bloom dates are among the most precisely recorded phenological indicators worldwide, with records from Kyoto spanning over 1,200 years. Compare cherry's lenticels to zelkova's -- same structure, different species.

## Procedure

1. Engage: Why might scientists care about the exact date cherry trees bloom each year?
2. Explore: Research how long-term cherry bloom records have been kept in Japan.
3. Explain: Introduce phenology as the study of seasonal timing in living things.
4. Art: Study Japanese hanami (cherry blossom viewing) traditions, then create a short ink-wash-style painting of a blossoming branch using only water and one color of paint.
5. Evaluate: Write a paragraph explaining what phenology is and why cherry bloom dates are a useful example of it.

## Discussion questions

What could a long historical bloom-date record reveal about a changing climate?

## Assessment

Paragraph correctly defines phenology and connects it to cherry bloom dates.

## Extension

If documenting in spring, log this tree's exact bloom date for a future comparison.

GRADE BAND 9-12

# Cherry: Evaluate phenology data as evidence in climate research.

**Standards connection:** NGSS HS-ESS3-5; National Core Arts Standards Proficient-level Creating/Responding (9-12)

**Materials:** Coloring sheet, research sources, ink or bold-outline art materials.

## Teacher background

Cherry bloom dates are among the most precisely recorded phenological indicators worldwide, with records from Kyoto spanning over 1,200 years. Earlier bloom dates are used as evidence of urban and global warming.

## Procedure

1. Engage: What makes a centuries-long bloom-date record such powerful climate evidence compared to modern temperature sensors alone?
2. Explore: Research how the Kyoto cherry bloom record has been used in peer-reviewed climate studies.
3. Explain: Evaluate the strengths and limitations of phenological records as climate evidence.
4. Art: Research ukiyo-e woodblock print conventions used by artists like Hokusai for depicting blossoming trees, then create your own print-style composition of this tree using bold outlines and flat color fields.
5. Evaluate: Submit a short written evaluation of phenology as climate evidence, plus the print-style artwork.

## Discussion questions

What other long-running human traditions (harvest festivals, planting dates) might double as unintentional climate records?

## Assessment

Rubric covering evidence evaluation depth and print artwork technique.

## Extension

If documenting in spring, log this tree's bloom date and compare it to historical Kyoto data trends.

## SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.